Package management

Linux has distribution-specific package management tools and formats for installing additional packages from the Internet or from a CD—like Red Hat RPM, while Debian and Ubuntu use DEB, and so on. The package management system in OpenBSD uses what is known as the *pkg_add* toolkit. Like everything else in OpenBSD, this is tightly integrated with the OS itself, and all the tools necessary for installing, maintaining, upgrading and removing packages are installed as part of the OS install. The package management system is written in Perl, which is pre-installed for OpenBSD. You can install packages from an install CD (which needs to be purchased) or off SSH, FTP or HTTP mirrors. It's simple to install a package. For example, *pkg_add socat* (as the super-user) will install the package *socat*.

Certain packages come with 'flavours', which determine whether they are built with Python or Lua support for X. For example, if you wish to install the multimedia application *mplayer*, then you could try *pkg_add -i mplayer*. The extra '-i' switch is important—it turns the installation into an interactive session, which will prompt you to choose a specific flavour of *mplayer*.

I normally install a certain minimum set of packages after a base install, which takes about 10 minutes, by issuing the following command:

```
# pkg_add -i mplayer vim socat colorls qemu windowmaker pidgin
firefox
```

Your choices may differ. In fact, if you end up using my LiveUSB project images, they come pre-built with a certain configuration, packages, etc.

Now I guess you are fairly familiar with OpenBSD and are ready to explore its potential to address basic and slightly advanced needs like running your own mail server, NAS backup system, and a very simple Web server. Doing this obviously requires domain knowledge and experience, which you can slowly acquire—or if you already have it, you can now do things in a new way using OpenBSD. First examine what problem you are trying to solve and then go about the implementation.

Mail server

A mail server sends and receives e-mail from various domains with multiple users, catering to the needs of local users and other e-mail servers. My company is in the business of creating OpenBSD-based products for email solutions, so I have personal experience of developing and maintaining a mail server for more than four years. If you are exposed to systems administration in your day job, then you probably already know that Internet email is very different from the simple local mail service that runs without any contact with the outside world. However, there is no need to worry. OpenBSD makes some things easy for us; while other things have to be learnt and only when you learn about them can you enjoy the benefits of the power of OpenBSD.

So you're now faced with the choice of which MTA (Mail Transfer Agent) to use. There are several popular options available, and though OpenBSD does not yet have its own implementation, it will very soon—Gilles Chehade and friends are hard at work on one called OpenSMTPD. I settled on Wietse Venema's Postfix MTA over alternatives like Sendmail, Qmail and Exim. Also, certain sites use Microsoft Exchange, which so far has never given me any pain. It is very odd—Windows machines often get virus-infected, sending out spam that can get your company's mail server's public IP blacklisted as a spam source. Other people on the Internet may abuse your mail server for purveying mail to unsuspecting third-party users. Yet, Microsoft Exchange, as an MTA, has not given me any trouble so far.

First, you obviously have to install Postfix, which is as simple as *pkg_add -i postfix*. It will throw up about eight choices, and you can select what you like. No big deal.

It is also very important to know that a mail server doesn't just send and receive mail using SMTP; it also has ancillary functions like distributing mail using IMAP and/or POP3, and acting as a webmail server. All this is part of the bargain. You will end up using the very brilliant Dovecot open source package for IMAP and POP3, which can also interact nicely with MS Outlook and Roundcube, a PHP webmail package. One of my customers tells me that using Roundcube is similar to using Outlook. It is that great and user-friendly. So to install, issue a simple… *pkg_add roundcube dovecot* and you're ready.

Roundcube

Easy going so far, right? Now, configuring Dovecot and Postfix is easy, but Roundcube presents certain difficulties, which we will look into. The first thing to do is identify the database Roundcube should use for storing mail—I normally use SQLite. Initialise a database with a script like what follows:

```
# sqlite -INIT SQL/sqlite.initial.sql sqlite.db
sqlite>.exit
#chmod o+rw sqlite.db
```

Then you have to configure Roundcube (edit *db.inc.php*) with the path to the initialised database file. You also have to configure the timezone, the domain, the default IMAP and SMTP server, the address-book (if using LDAP), enable the preview pane and so on, in the configuration file *main.inc.php*.

Following this, the kind souls who developed Roundcube have given an installer that tests the installation in many ways. Before you get to that, however, you have to enable PHP in Apache, and also add a section with directives for allowing the *.htaccess* magic within the *webmail* directory. Another very important thing to remember is that Roundcube will only run in a non-*chroot*-ed Apache server. Be very careful about that, since by default OpenBSD runs Apache in a *chroot* environment, jailed within */var/www*. Once all this is done, and the Roundcube installer is moved out of the way, you can access and use webmail. But you have to do some configuration tweaks to get there, which we tackle later.